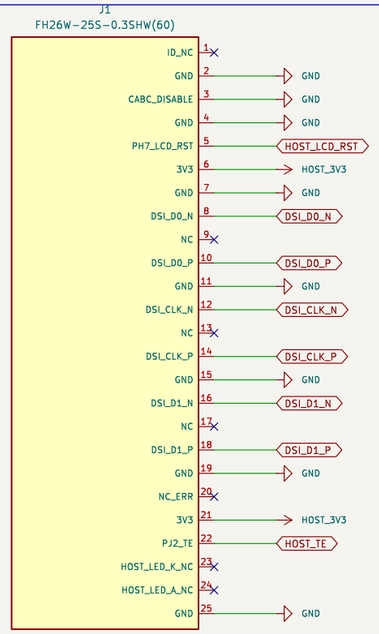
01 / STM32F469I-DISCO > DISEN DS070HSD39T-132

MIPI DSI / LCD interface | Host CN10: UM1932 Rev 5 tables 6–7 | Panel: DISEN A0 preliminary

01 / STM32 HOST – CN10



25-pin FPC / 0.30 mm pitch

HOST CONNECTION NOTES

Remove original LCD and touch flexes.
Use FH26-compatible 0.20 mm FPC ends.
Verify actual pin 1 and 1:1 continuity.

J1.23 / J1.24: host LED outputs unused.
J1.3 = GND disables default CABC control.
MB1189-C01: R117 fitted / R119 absent.
Confirm actual host revision and population.

J-SR1 / LAB PROTOTYPE – Electrical schematic review; hardware operation is not certified.
Panel timing/current, host flex pin order and OVP transient qualification remain open; see report.

02 / MIPI DSI – DIRECT PAIRS

NET / PAIR	HOST J1	LCD J2
DSI_D0_P	10	16
DSI_D0_N	8	17
DSI_D1_P	18	20
DSI_D1_N	16	21
DSI_CLK_P	14	24
DSI_CLK_N	12	25

03 / RESET & FRAME SYNCHRONIZATION



No DSI series resistors or AC coupling.

05 / POWER, TOUCH & BACKLIGHT SHEETS

Logic power & touch

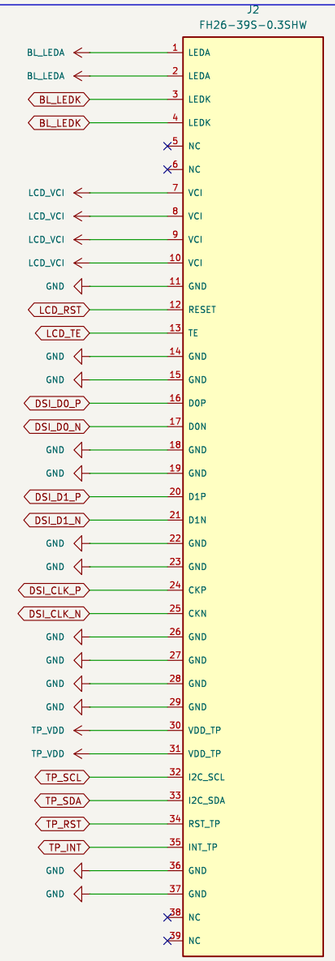
File: Logic_Touch.kicad_sch

Backlight / external power

File: Backlight.kicad_sch

Logic: host 3.3 V only. Boost: separate regulated 5 V.
All blocks share GND. Identical net labels are connected.

04 / DISEN LCD – 39-PIN FPC



0.30 mm pitch / verify contact side & pin 1

DSI: 2 data lanes + clock, direct DC coupling.
100 ohm nominal differential; continuous GND.
OTA7290B / ILI2130 are panel devices.
J2 corrected to FH26; verify physical flex mating.

H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole

Prototype | See review report
Yakup Turkan

Sheet: /
File: STM32F469_DISEN_RevJ_Prep1.kicad_sch

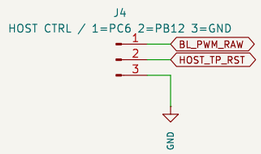
Title: STM32F469 / DISEN – MIPI & LCD interface

Size: A3	Date: 2026-10-22	**Rev: J-SR1**
KiCad E.D.A. 9.0.2		Id: 1/3

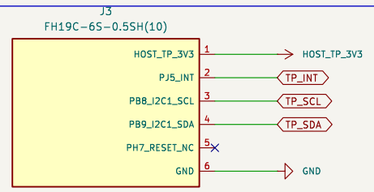
02 / LOGIC POWER, TOUCH & HOST CONTROL

3.3 V logic domains | LCD and touch reset kept independent | Test points grouped by function

01 / CN12 THREE-WIRE CONTROL HARNESS



02 / TOUCH HOST – CN2



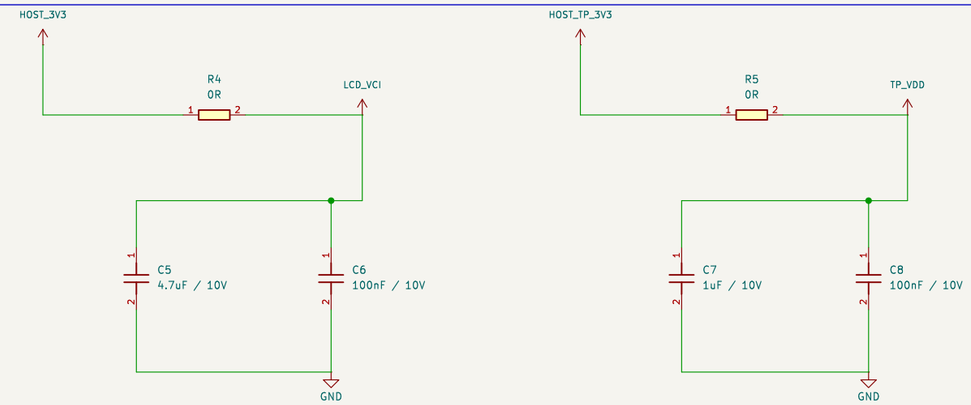
HOST CN2 -> J3 only (6-pin FPC, 0.50 mm).
1=3V3; 2=INT; 3=SCL; 4=SDA; 5=NC; 6=GND.
Do NOT connect the panel 6-pin touch flex here: pinout differs.
Panel touch is connected through J2 pins 30-35.

INTERFACE CONVENTIONS

Three-wire J4 harness (not a CN12 1:1 socket):
J4.1 -> CN12.6 / PC6 PWM
J4.2 -> CN12.12 / PB12 touch RESET
J4.3 -> CN12.2 / GND

J3 touch FPC: pinout verified for MB1189-C01.
Both logic rails are supplied by host 3.3 V.

03 / LOGIC SUPPLIES & DECOUPLING

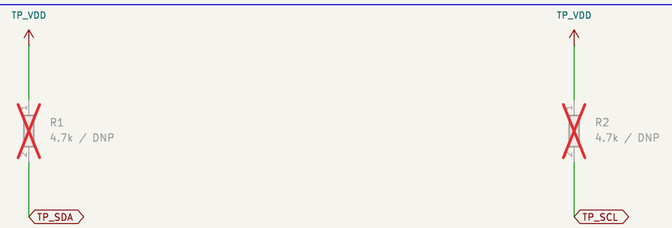


LCD_VCI / host LCD 3.3 V

TP_VDD / host touch 3.3 V

Keep 5 V away from both logic rails. Current budget and voltage tolerance require validation.

04 / TOUCH I2C – OPTIONAL PULL-UPS

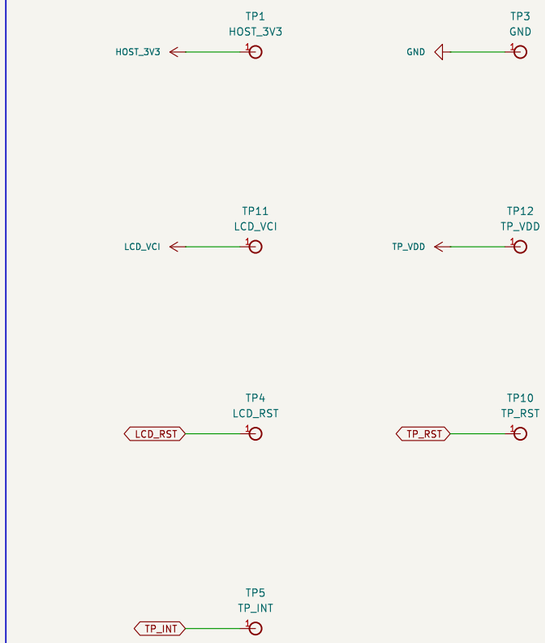


MB1189-C01: PB8/PB9 = I2C1; R140/R142 = 1.5k pull-ups.
R1/R2 stay DNP. R36 = 4.7k pulls PJ5 / INT high.
Confirm actual host revision; measure bus rise time with the FPC.

VALIDATION NOTES – J-SR1

LCD VCI: 3.0-3.6 V; touch: 3.3 V nominal.
Drawing specifies touch 2.8-3.3 V: confirm tolerance.
LCD and touch currents are TBD: measure host load.
RESX/ILI2130 timing not specified sufficiently in panel PDF.

05 / LOGIC TEST POINTS



Supply declarations



Prototype | See review report
Yakup Turkan

Sheet: /Logic power & touch/
File: Logic_Touch.kicad_sch

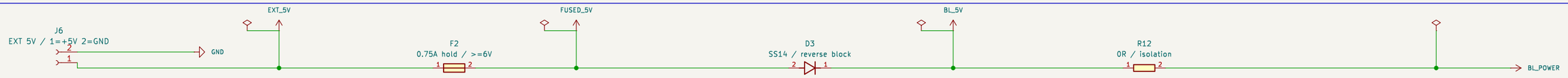
Title: STM32F469 / DISEN – Logic power & touch

Size: A3	Date: 2026-10-22	**Rev: J-SR1**
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03 / EXTERNAL 5 V & TPS61165 BACKLIGHT

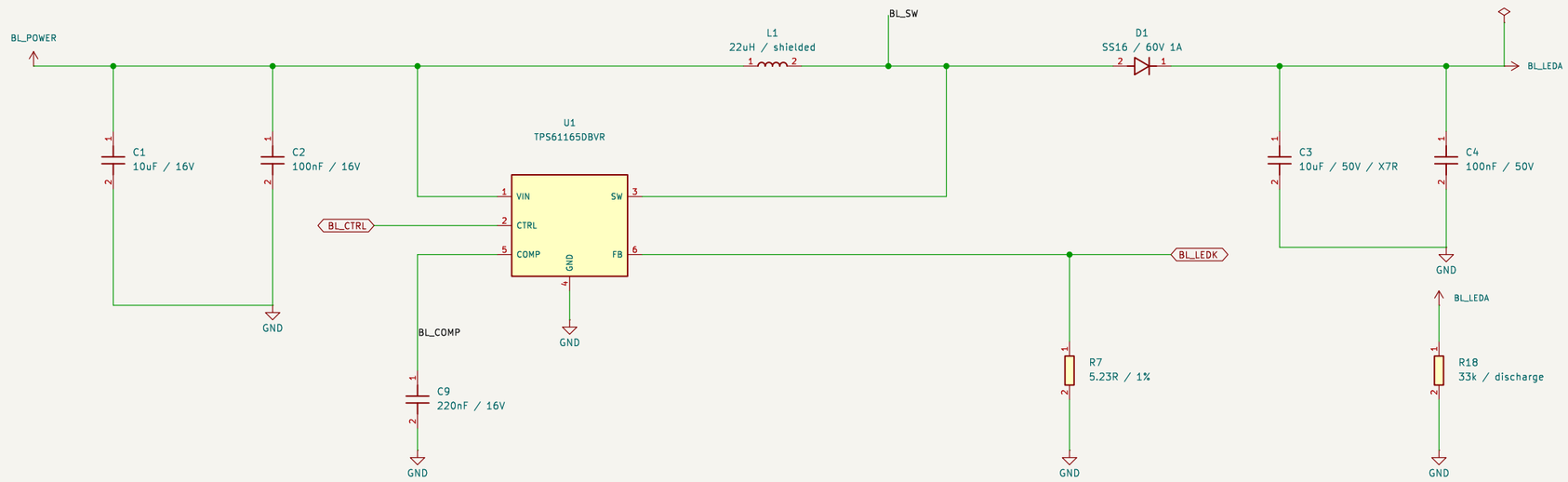
LED load: 21.6–27.2 V at 40 mA | Separate regulated boost supply | Common GND with F469

01 / INPUT POWER – FUSE, REVERSE BLOCK & ISOLATION



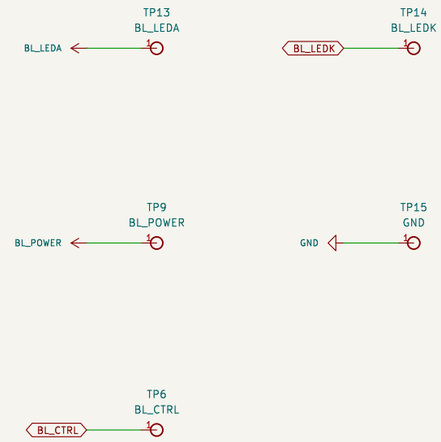
Regulated 5 V via J6: pin 1 = +5 V; pin 2 = GND. Host +5 V and LEDA/LEDK are NOT connected.

02 / BOOST POWER STAGE & LED CURRENT RETURN

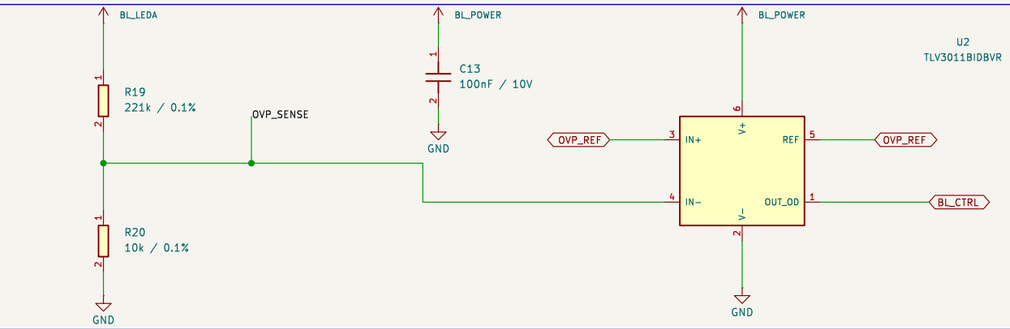


I_LED = 0.200 / 5.23 = 38.24 mA nominal; worst-case full-scale = 39.40 mA (TI VFB + R7 tolerance).

03 / BACKLIGHT TEST POINTS

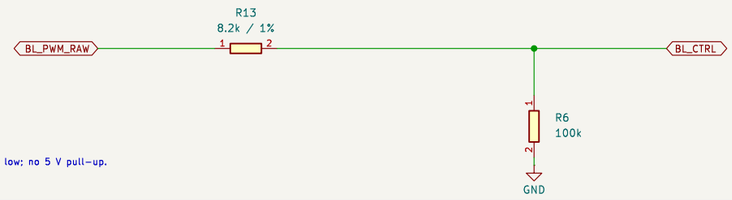


04 / OVP SENSE & COMPARATOR SHUTDOWN



OVP nominal = 28.690 V; U2 must be TLV3011B (POR + fail-safe + internal hysteresis).
POR may leave OUT Hi-Z for up to 1.9 ms. Hold PWM LOW through rail transitions; wait >=5 ms after rails valid.
TPS61165 CTRL filter = 180 us; shutdown LOW >2.5 ms. Target PWM 20 kHz; R13=8.2k limits ACK current.
FH26 family is rated 30 V: verify open-load peaks <30 V. OVP is NOT a hard clamp; see engineering report.

05 / PWM CONTROL & DEFAULT-OFF



Open-drain U2 pulls CTRL low; no 5 V pull-up.

Prototype | See review report
Yakup Turkan
Sheet: /Backlight / external power/
File: Backlight.kicad_sch

Title: STM32F469 / DISEN – External 5 V & backlight

Size: A3	Date: 2026-10-22	**Rev: J-SR1**
KiCad E.D.A. 9.0.2		Id: 3/3